How Much Does Scoring Change the Story?

Psychometric Stability in the Heart and Flower Task Through a Multiverse Lens

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Abstract

The *Heart and Flower* task is a widely used tool for assessing children's executive functioning. Depending on the study, performance may be indexed by accuracy, reaction times, difference scores, composite measures, or joint accuracy–RT formulations reflecting the lack of a common scoring method. This plurality may foster questionable measurement practices, which in turn can yield different conclusions hindering comparability of results across studies.

We address this issue with a psychometrically oriented *multiverse analysis*. Starting from trial-level task data, we examined associations with age as an indicator of construct validity while systematically varying key analytic decisions. The multiverse included alternative subject inclusion rules, different RT summaries, congruent-referenced difference scores, inverse efficiency metrics, standardized accuracy–RT composites, and interaction-based scoring approaches. This generated a structured space of plausible scoring specifications rather than a single privileged index.

For each specification, age effects were estimated with subject-level linear models and generalized additive models, and with trial-level mixed-effects models including subject random intercepts and condition slopes, alongside GAMs with smooth age functions and subject random effects. To summarize robustness across the analytic space, we used specification curves, volcano plots, and model-based predictive visualizations.

Rather than asking whether the task "works" under one scoring rule, this study asks which conclusions survive across many defensible choices. The goal is to identify indices that show stable age-related gradients, distinguish them from specification-sensitive effects, and provide a transparent framework for evaluating the psychometric robustness of executive function measures when multiple scoring approaches are simultaneously plausible.

Keywords: Heart and Flower task; executive functions; multiverse analysis; psychometric robustness; scoring; construct validity